MOBILE APP USER ENGAGEMENT & CHURN ANALYSIS

Final Business Analytics Report

A data-driven review of user engagement, subscription mix, churn-risk profiling, product usage and retention opportunities.

Dataset	Mobile App User Engagement
Source records	20,000
Distinct User IDs	2,000
Source fields	15
Derived metric	Engagement Score
Login period	01 Jan 2023 to 01 Jun 2025
Analysis layer	Python notebook + SQL business queries

Purpose
Translate user-level behavioral data into clear findings and practical actions for engagement, retention and product decision-making.

Prepared from the supplied CSV, Python analysis notebook and SQL analysis layer.

1. Executive Summary

This analysis covers **20,000 source records** spanning 01 Jan 2023 to 01 Jun 2025. The raw file contains **2,000 distinct User IDs**, so the 20,000-row dataset should be described as records rather than 20,000 unique users. The analysis reproduces the supplied notebook's engagement calculation and extends it with a repeatable SQL layer for business questions.

Overall usage is stable and broadly consistent across demographic and platform segments. Average sessions are **7.47 per day**, average session duration is **30.45 minutes**, the mean Engagement Score is **245.54**, and mean Churn Risk Score is **0.497**.

The subscription base is balanced across Free, Trial and Premium. Churn-risk averages are almost identical between these groups, which means subscription status alone is not a strong discriminator of risk in this dataset.

The strongest analytical opportunity is behavioral segmentation: identify users who combine elevated risk with relatively low engagement, then test targeted retention interventions. This is risk profiling, not a measured churn study, because the source file does not contain an observed churn or cancellation outcome.

Headline findings

Metric	Result	Business reading
Records	20,000	Large analytical sample
Distinct User IDs	2,000	Repeated IDs exist in raw data; treat rows as records
Avg sessions/day	7.47	Consistent usage frequency
Avg session duration	30.45 min	Substantial session depth
Avg Engagement Score	245.54	Composite behavioral index
Avg Churn Risk Score	0.497	Near midpoint of 0-1 scale
Avg User Rating	2.99/5	Moderate satisfaction level

Decision focus: use SQL and dashboard views to move from descriptive reporting toward repeatable behavioral segmentation, retention testing and outcome validation.

2. Data Overview & Methodology

The source CSV contains **20,000 rows and 15 source fields**. The supplied notebook loads the data with Pandas, creates a composite Engagement Score, converts Last Login Date to datetime, creates age groups and Login Month, and produces grouped summaries for visualization.

Source fields

Category	Fields
User identity	User ID, Gender, Age
Geography & platform	Country, Device Type, App Version
Engagement	Sessions Per Day, Avg Session Duration Min, Screens Viewed
Messaging	Push Notifications Clicked
Monetization	In App Purchases, Subscription Status
Risk & satisfaction	Churn Risk Score, User Rating
Time	Last Login Date

Engagement Score

The notebook defines Engagement Score as: **(Sessions Per Day × Avg Session Duration Min) + (Screens Viewed × 0.5) + (In App Purchases × 2)**. This is a custom composite index. It is useful for ranking and segmentation within this dataset, but it is not a standard industry metric.

Age segmentation

Age is grouped into **<18, 18-24, 25-34, 35-44, 45-54, 55-64 and 65+**, matching the notebook logic.

Data-quality observation

The raw CSV has **0 missing values** across the 15 source fields. A separate identifier check shows 2,000 distinct User IDs across 20,000 rows. This indicates that the file contains repeated User IDs. For this reason, monthly activity and segment counts in this report are record-based unless explicitly stated otherwise.

SQL analysis layer

A MySQL 8+ SQL file was created from the same source logic. It includes a raw table, CSV load pattern, data-quality checks, an analytics view with Engagement Score and Age Group, plus business queries for KPIs, subscriptions, churn risk, app versions, devices, countries, monthly logins and behavioral segments.

3. Core Engagement Profile

Average sessions per day are **7.47**, average session duration is **30.45 minutes**, average screens viewed are **26.99**, push-notification clicks average **4.46**, and in-app purchases average **2.00**.

Figure 1. Distribution of the six core engagement, risk and satisfaction measures.

Engagement Score interpretation

The Engagement Score has a mean of **245.54**, median of **191.94**, standard deviation of **191.26**, and maximum of **868.94**. The mean being above the median indicates a right-skewed distribution, with a smaller group of highly engaged records pulling the average upward.

The score is strongly associated with average session duration (r=0.672) and sessions per day (r=0.650). That relationship is expected because both variables are explicitly included in the formula. The score should therefore be treated as an index, not as an independent validation of engagement.

4. Segment-Level Findings

Age group engagement

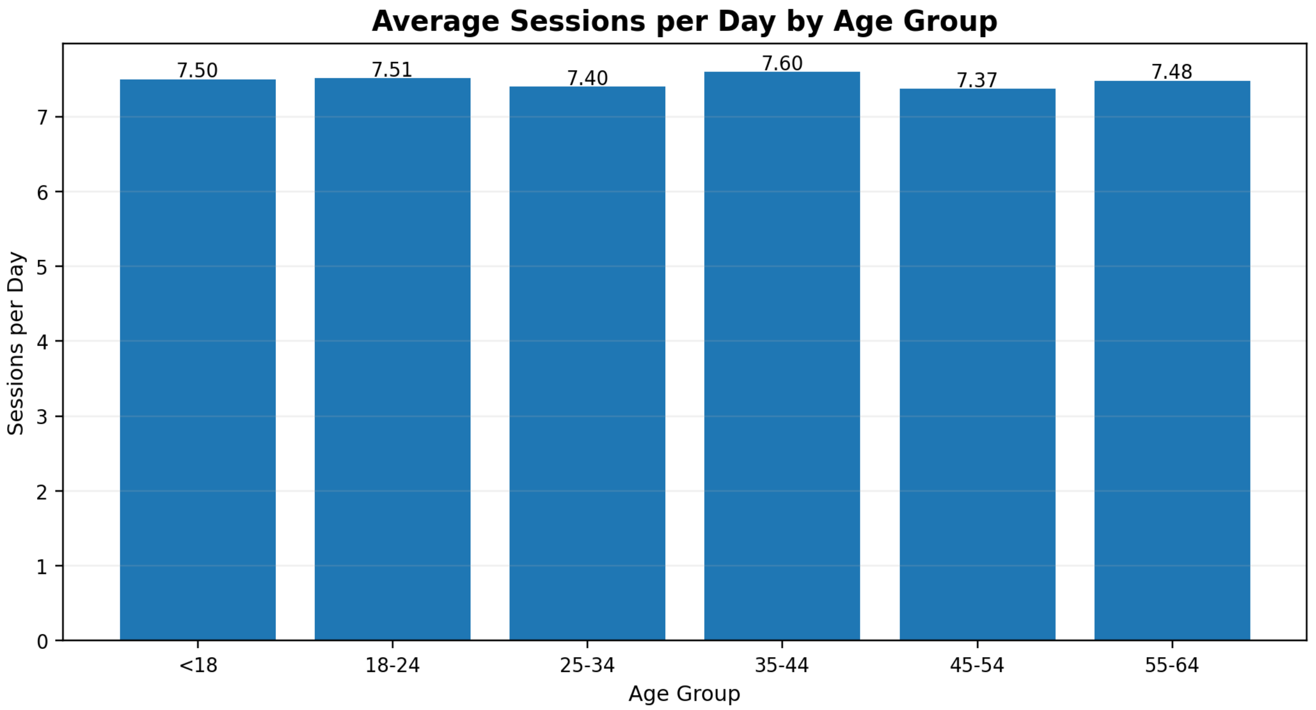

Figure 2. Average sessions per day by age group. The highest observed average is the **35-44** group at **7.60**, while the **45-54** group is lowest at **7.37**. The spread is small, so age is not a strong differentiator of session frequency in this sample.

Country engagement

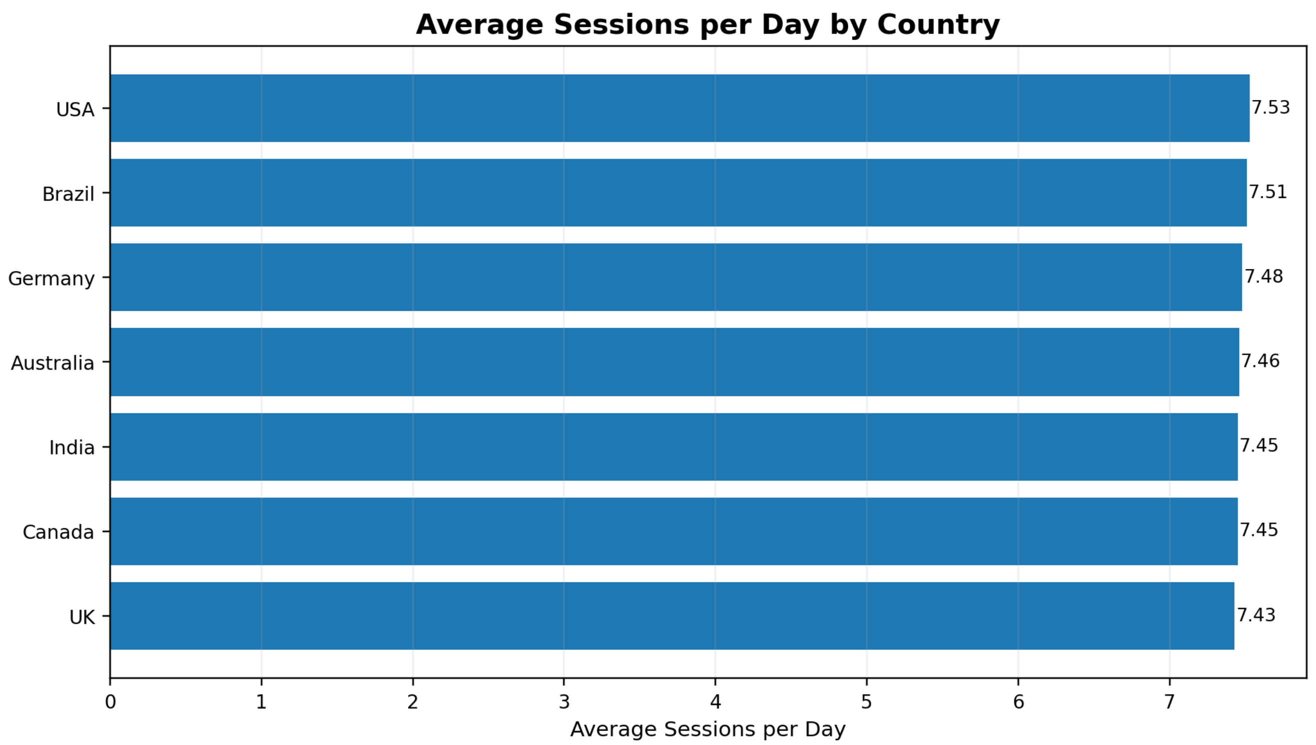

Figure 3. Average sessions per day by country. The USA is highest at **7.53**, while the UK is lowest at **7.43**. The difference is only **0.10 sessions per day**, indicating modest geographic variation.

5. Subscription Mix & Churn-Risk Profile

The subscription mix is balanced: Free has **6,653 records (33.3%)**, Trial has **6,580 (32.9%)**, and Premium has **6,767 (33.8%)**.

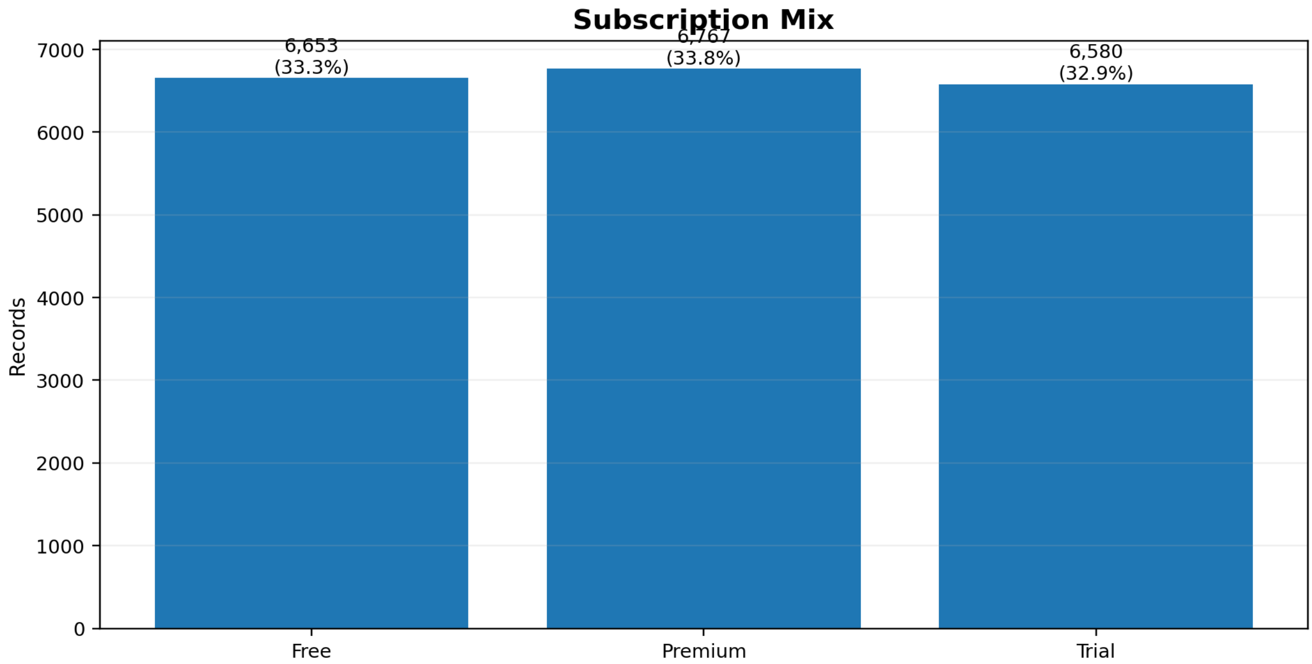

Figure 4. Subscription mix across the source records.

Subscription	Records	Share	Avg Risk	Avg Engagement	Avg Sessions
Free	6,653	33.3%	0.499	244.32	7.42
Trial	6,580	32.9%	0.493	244.28	7.53
Premium	6,767	33.8%	0.499	247.96	7.47

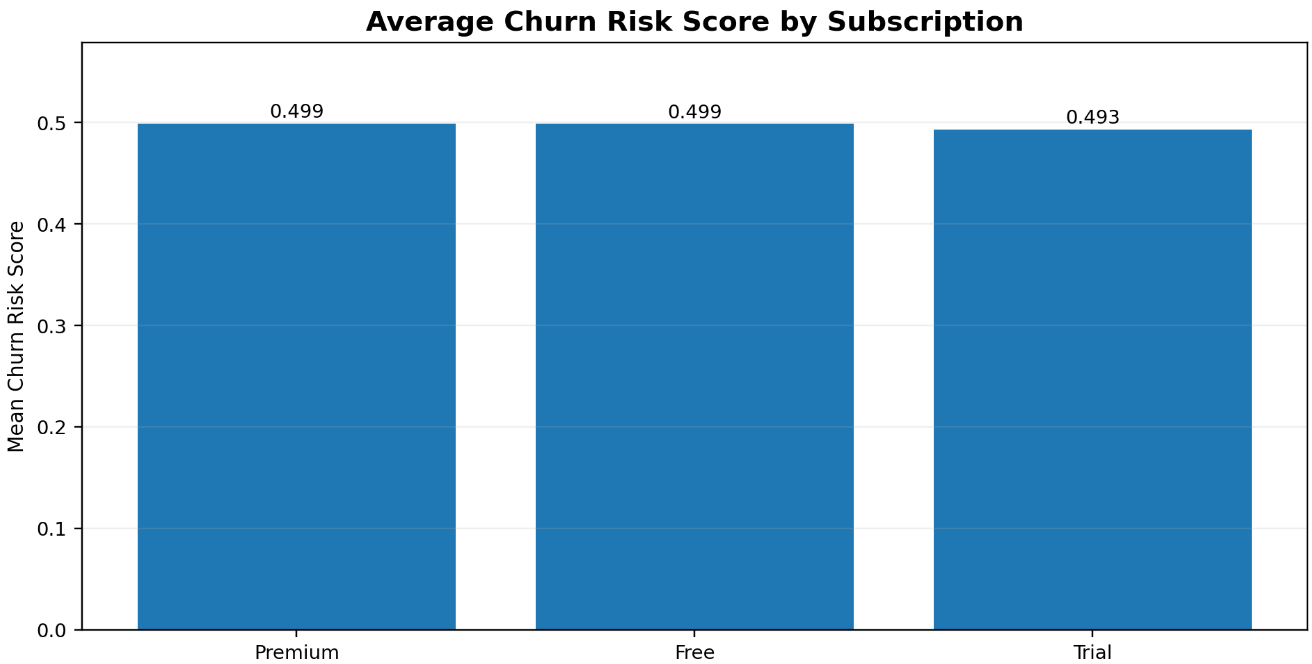

Figure 5. Mean Churn Risk Score by subscription status.

Average risk is nearly identical: Free 0.499, Trial 0.493, and Premium 0.499. Subscription status alone therefore provides little separation of the supplied risk score.

Interpretation: the Churn Risk Score is a supplied risk indicator. There is no observed churn, cancellation or downgrade label in the source data, so this analysis cannot report a measured churn rate or establish churn drivers.

6. Behavioral Retention Segmentation

A more actionable retention view combines risk and engagement. Using the dataset mean as the split point (risk threshold 0.497; engagement threshold 245.54), each record is assigned to one of four behavioral segments.

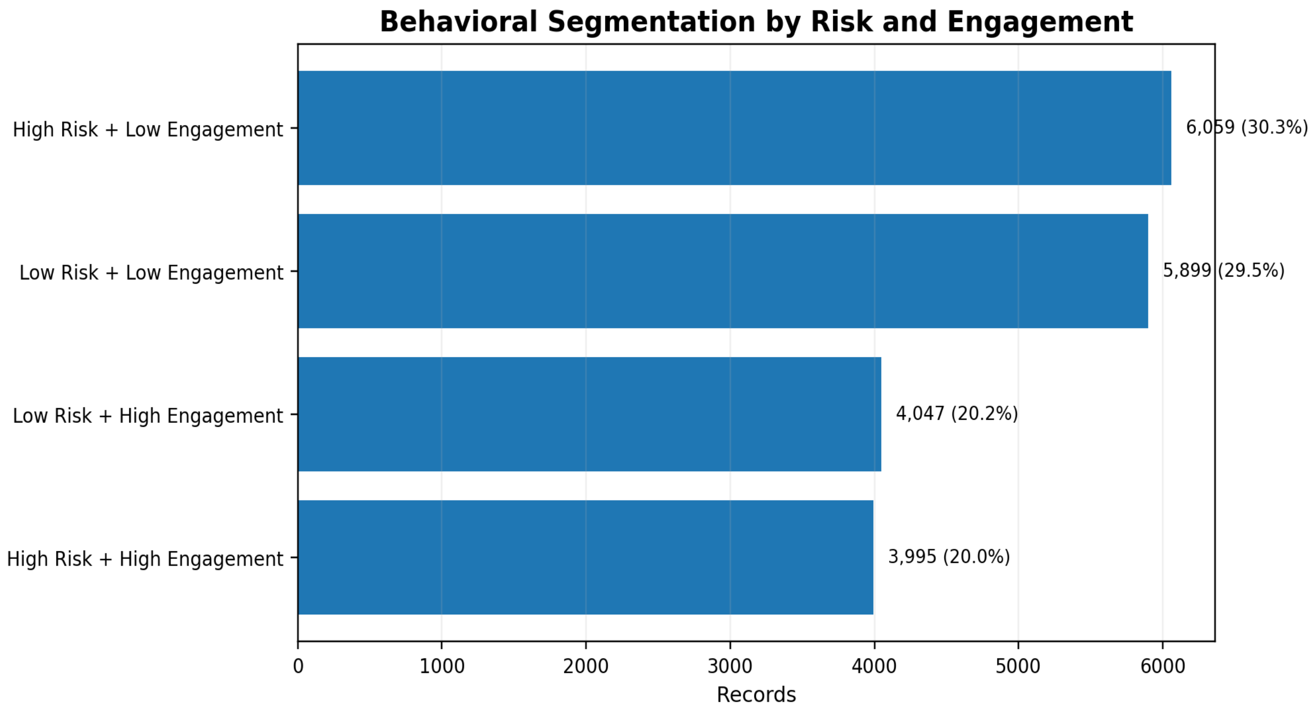

Figure 6. Four-way behavioral segmentation using supplied risk and derived engagement.

Segment	Records	Share	Avg Risk	Avg Engagement
High Risk + Low Engagement	6,059	30.3%	0.746	114.93
Low Risk + Low Engagement	5,899	29.5%	0.245	114.50
Low Risk + High Engagement	4,047	20.2%	0.247	439.26
High Risk + High Engagement	3,995	20.0%	0.744	440.88

The highest-priority candidate for a retention experiment is **High Risk + Low Engagement**. This segment combines the two signals that are most useful for intervention design: a high supplied risk score and relatively weak behavioral engagement.

Recommended actions include targeted re-engagement, product education, personalized messaging and incentive testing. These should be treated as experiments and evaluated against subsequent activity or real retention outcomes, rather than assumed to work from this descriptive analysis alone.

7. Product Usage & Time Trends

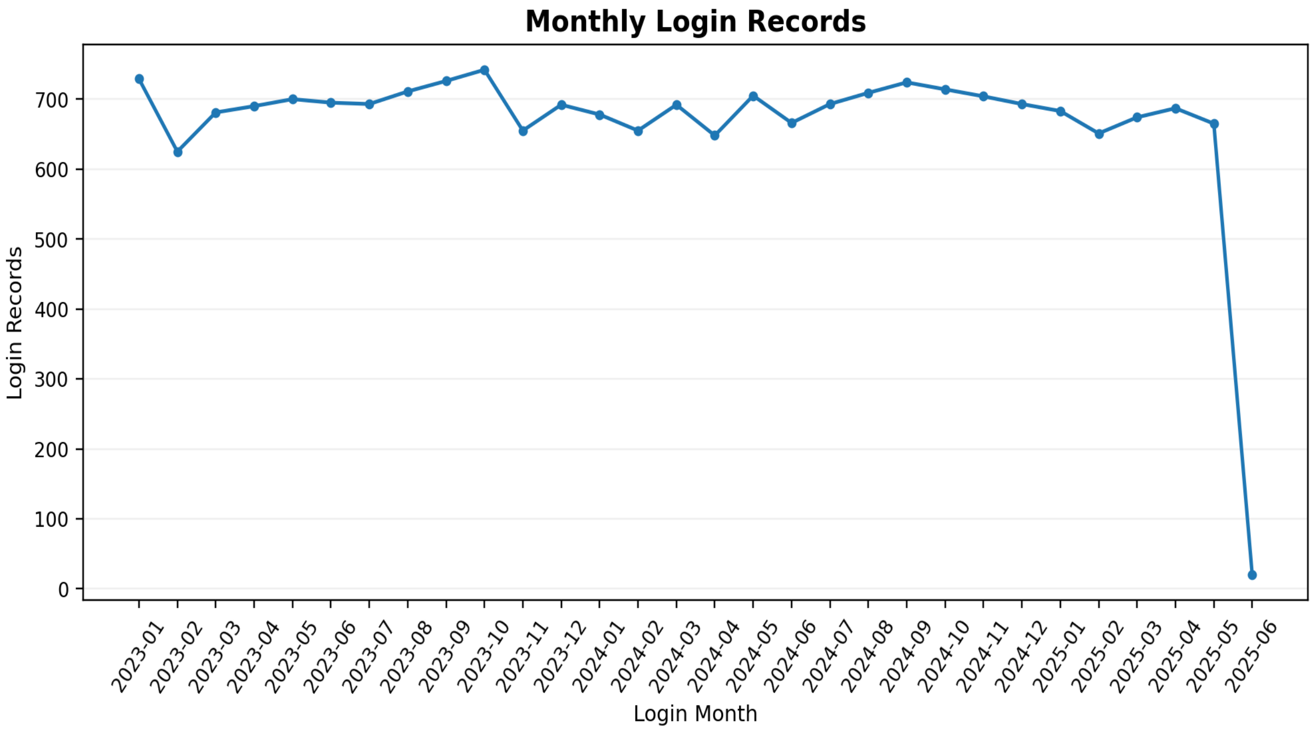

Figure 7. Monthly login records based on Last Login Date.

Login records are broadly stable across most full months. The highest monthly count is **742** in **2023-10**. The final observed month, **2025-06**, contains only **20** records, far below the hundreds seen in preceding months.

Important: the final month should not be interpreted as a product collapse without confirming data completeness. It may be a partial extract or reporting cutoff. Also, these are record counts by last-login month, not guaranteed unique monthly active users.

Push-notification interaction

Users average **4.46** push-notification clicks, with a median of **4**. The distribution spans 0 to 9 clicks.

The descriptive distribution alone cannot establish notification effectiveness. A stronger design would connect notification exposure to subsequent engagement, inactivity or retention outcomes.

8. Device & App-Version Analysis

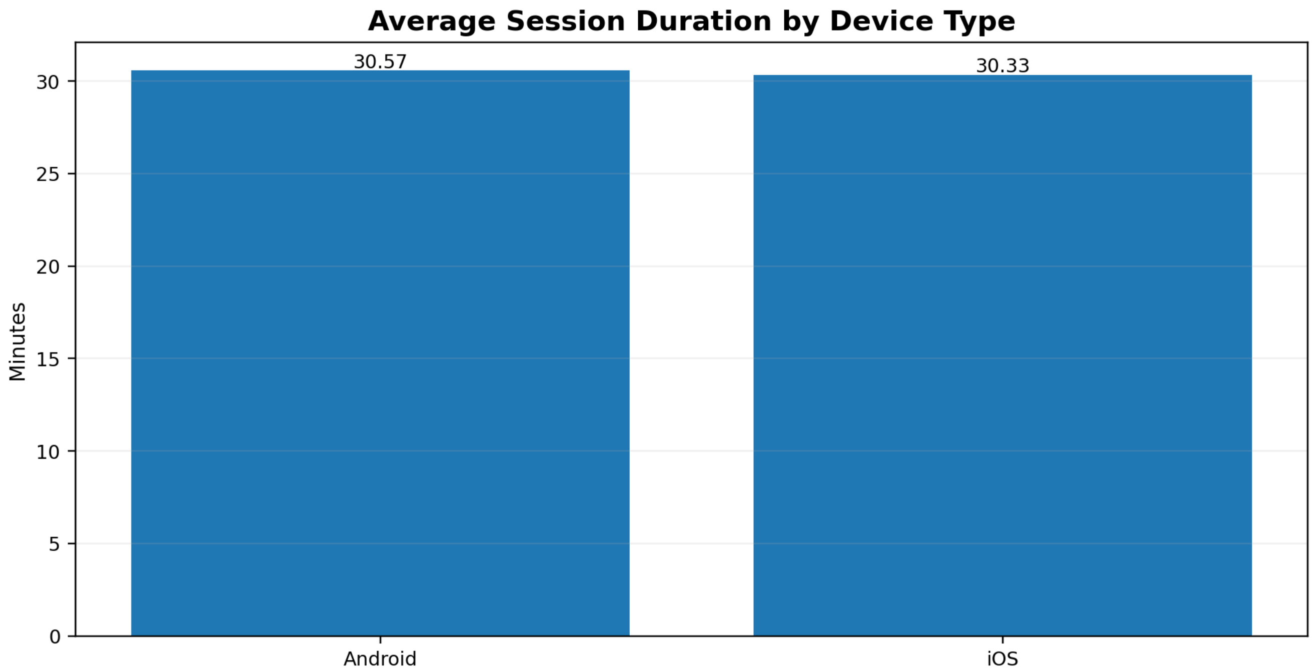

Figure 8. Average session duration by device type.

Android records average **30.57 minutes** per session compared with **30.33** on iOS, a difference of only **0.24 minutes**. This suggests broadly similar session depth across the two platforms.

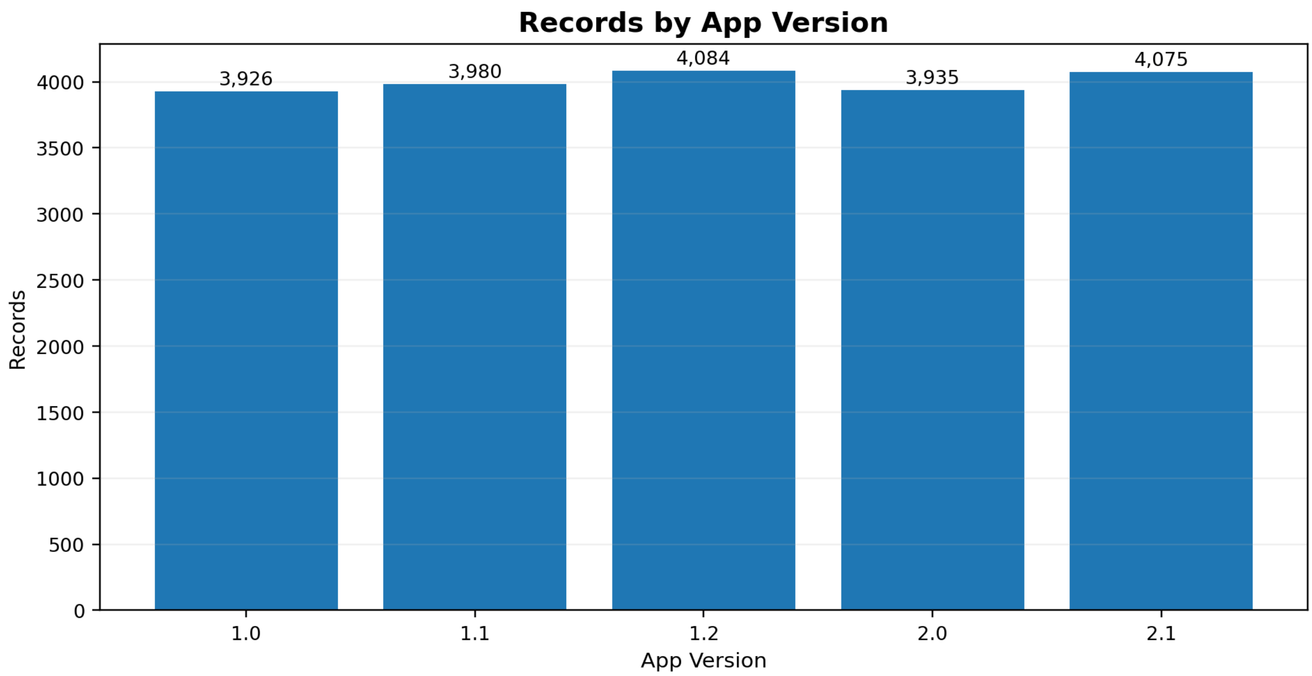

Figure 9. Source-record distribution by app version.

Version	Records	Avg Sessions	Avg Duration	Avg Engagement	Avg Risk	Avg Rating
1.0	3,926	7.44	30.58	244.03	0.495	2.97
1.1	3,980	7.48	30.23	245.22	0.497	2.98
1.2	4,084	7.48	30.40	245.81	0.493	3.00
2.0	3,935	7.41	30.39	243.39	0.501	3.02

Version	Records	Avg Sessions	Avg Duration	Avg Engagement	Avg Risk	Avg Rating
2.1	4,075	7.55	30.65	249.10	0.499	2.99

Version 1.2 has the largest record base at 4,084, followed by version 2.1 at 4,075. Adoption alone is not enough to judge release performance, so engagement, rating and risk should be compared across versions.

9. Satisfaction & Relationship Analysis

The average user rating is **2.99/5**, with a median of **3.0**. This indicates a moderate satisfaction level in the source records.

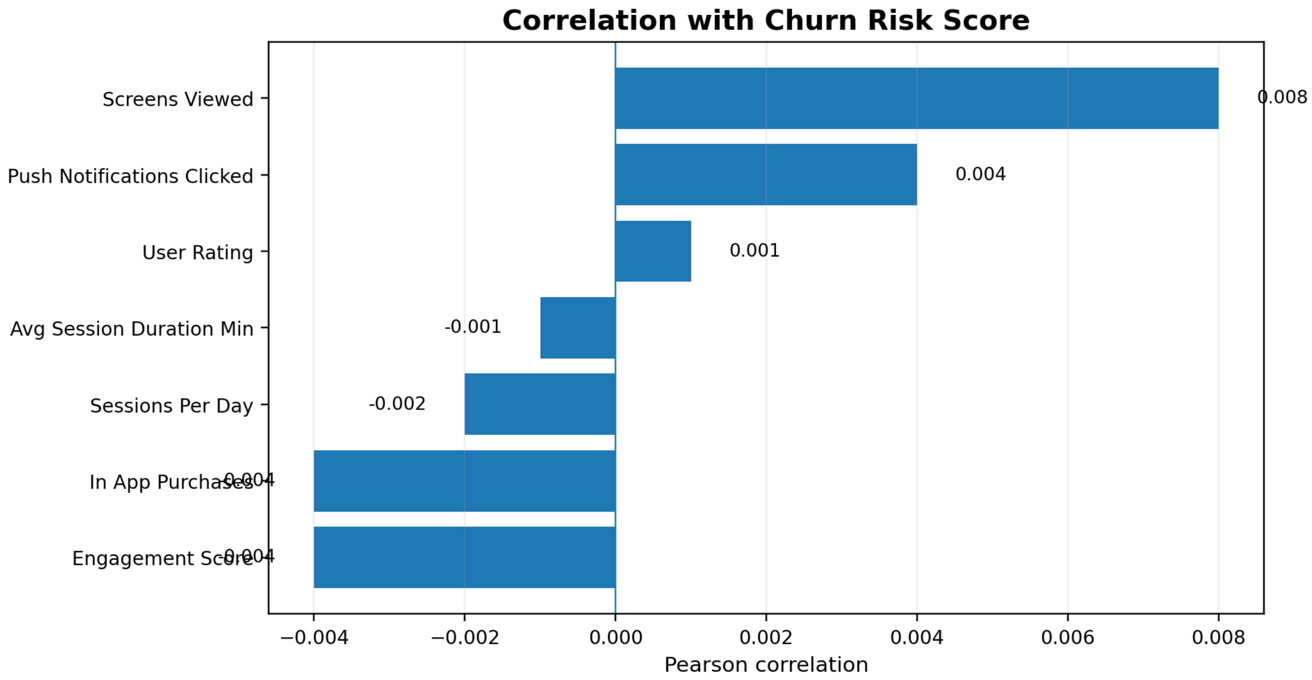

Figure 10. Pearson correlation of measured variables with the supplied Churn Risk Score.

Variable	Correlation with Churn Risk
Engagement Score	-0.004
In App Purchases	-0.004
Sessions Per Day	-0.002
Avg Session Duration Min	-0.001
User Rating	0.001
Push Notifications Clicked	0.004
Screens Viewed	0.008

Every tested variable has a correlation close to zero with Churn Risk Score. The largest absolute relationship is Screens Viewed at **0.008**, which remains negligible in linear terms.

This suggests the supplied risk score is not linearly explained by any single measured behavior in this dataset. It does not prove that the risk score is invalid, because correlation is not a predictive validation test and does not establish causation.

Engagement Score is correlated with session duration (r=0.672) and sessions per day (r=0.650). That is expected because these variables are part of the score's construction.

10. SQL Analysis Layer

The project now includes a MySQL 8+ SQL layer designed to make the analysis repeatable and easier to connect to a BI dashboard. The SQL file recreates the notebook's derived logic and turns the analysis into reusable business queries.

SQL component	Purpose
Raw table	Stores the 15 source fields from the CSV
Data-quality checks	Record count, distinct IDs and missing-value checks
Analytics view	Adds Engagement Score, Age Group and Login Month
KPI queries	Core averages, ranges and summary metrics
Segment queries	Age, country, subscription and behavioral segments
Risk queries	Risk distribution and subscription comparisons
Product queries	App version, device and notification performance
Time queries	Monthly login activity and latest-period checks
Dashboard extracts	Country/subscription and device/version summaries

Example business questions now answerable in SQL

- Which subscription group has the highest supplied churn risk?
- Which countries have the highest average session frequency?
- Which app versions have the strongest engagement and ratings?
- How does session duration differ by device?
- How many records fall into High Risk + Low Engagement?
- Which month has the highest login-record volume?
- How does notification interaction relate to engagement and risk descriptively?

The SQL layer should be paired with an observed outcome table in the next iteration. That would allow real churn, inactivity, conversion and retention analysis instead of relying only on supplied risk scores.

11. Business Recommendations

1. Prioritize High Risk + Low Engagement

Create a retention audience from users who combine elevated supplied risk with below-average engagement. Test targeted re-engagement, product education and personalized messaging.

2. Improve Trial-to-Premium analysis

Trial represents 32.9% of the records (6,580). The current file does not contain conversion history, so a true conversion rate cannot be calculated. Add trial start/end and conversion events, then compare converters and non-converters by behavior.

3. Validate the churn-risk score

Link the supplied risk score to actual inactivity, cancellation, downgrade or renewal outcomes. If labels become available, evaluate AUC, precision, recall, calibration and lift by risk band.

4. Resolve the June 2025 data issue

The final month has only 20 records versus hundreds in preceding months. Confirm whether this is a partial extract or reporting cutoff before using it for trend decisions.

5. Monitor app-version performance

Track engagement, ratings and supplied risk by version, not adoption alone. This can highlight releases associated with better or worse user experiences.

6. Test notification effectiveness

Use notification exposure and subsequent behavior to test whether targeted messaging increases engagement or retention. Descriptive click counts cannot establish effectiveness.

7. Build the BI layer

Create a Power BI dashboard with filters for country, device, subscription, age group and app version. Connect it to the SQL analytics view for repeatable reporting.

Recommended priority order: validate the data grain and outcomes first, then build behavioral retention segments, followed by trial conversion and notification experiments.

12. Limitations & Next-Phase Analysis

Current limitations

• The raw file contains 20,000 records but only 2,000 distinct User IDs, so record-level counts should not automatically be described as unique users.
• No observed churn/cancellation label is included, so actual churn cannot be measured.
• Engagement Score uses custom weights and has not been validated against a business outcome.
• Monthly login counts are record counts by Last Login Date, not necessarily unique active users per month.
• June 2025 appears unusually low and may be a partial period.
• Descriptive comparisons and correlations do not establish causality.
• The dataset does not include cohort retention, trial conversion, revenue or lifetime value.

Recommended next phase

1. Confirm the intended grain of the source data and whether User ID should be unique.
2. Add observed churn, inactivity and subscription-event labels.
3. Build retention cohorts and monthly active-user metrics using distinct users where appropriate.
4. Add Trial → Premium conversion events and analyze conversion by engagement segment.
5. Connect the SQL layer to a Power BI dashboard.
6. Validate the supplied risk score against real outcomes.
7. Run controlled retention and notification experiments.

Conclusion

The analysis shows a broadly consistent engagement profile across countries, age groups and devices, alongside a balanced subscription mix. The supplied Churn Risk Score does not differ materially by subscription and has almost no linear relationship with the measured behavioral variables. The most useful path forward is therefore not to label current users as churners, but to use the supplied score as one input to behavioral segmentation and then validate those segments against real downstream outcomes.

Appendix A. Core Metrics

Metric	Mean	Median	Min	Max
Sessions Per Day	7.47	7.00	1.00	14.00
Avg Session Duration Min	30.45	30.52	1.00	60.00
Screens Viewed	26.99	27.00	5.00	49.00
Push Notifications Clicked	4.46	4.00	0.00	9.00
In App Purchases	2.00	2.00	0.00	4.00
Engagement Score	245.54	191.94	4.13	868.94
Churn Risk Score	0.50	0.50	0.00	1.00
User Rating	2.99	3.00	1.00	5.00

Appendix B. Key Segment Tables

Country	Avg Sessions/Day
USA	7.53
Brazil	7.51
Germany	7.48
Australia	7.46
Canada	7.45
India	7.45
UK	7.43

Appendix C. Source & Deliverables

Source files used: Mobile App User Engagement CSV, supplied Python analysis notebook and prior report. Deliverables from this revision: a polished final PDF report, an editable DOCX version, and a MySQL 8+ SQL analysis file.

The SQL file is the repeatable analysis layer. The report summarizes the notebook outputs and the SQL-ready business questions without presenting descriptive risk as actual churn.